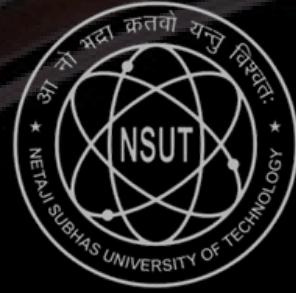


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INDIA INNOVATES 2026

Team Name: Team AGNI

**Members Name and
Affiliation:**

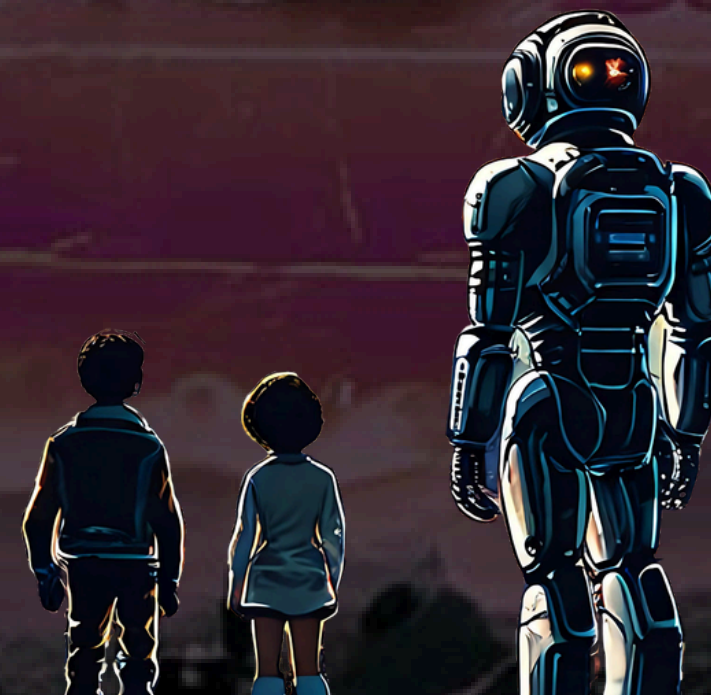
Pranav Nimje

Rishi Shahu

Bhagyesh Dedmuthe

Ketan Lohakare

Yashraj Talegaonkar





PROBLEM STATEMENT



India faces severe delays in emergency response due to traffic congestion and poor system coordination. Every year, the country records ~4.6 lakh road accidents and over 1.68 lakh deaths, with 50–60% of victims dying because they do not receive medical care within the Golden Hour (60 minutes). Ambulances often lose 15–25 minutes in traffic signals and congestion, accidents on highways are reported late, and ambulances sometimes reach hospitals only to find no available beds. In many highway areas, poor or no internet connectivity prevents drivers from contacting services like NHAI, even in emergencies such as a vehicle breakdown. Current systems rely heavily on centralized internet networks that fail in low-connectivity areas. Therefore, there is a need for an integrated intelligent system that can detect accidents automatically, enable vehicle communication through RSU hopping even without internet, optimize traffic signals using AI without changing the existing traffic system, and provide real-time hospital availability so ambulances can reach the nearest hospital faster and save lives.



R.A.A.M

(Road Accident Assistance Mechanism)

Satellite-Independent Mesh: Uses MEC (Multi-access Edge Computing) and roadside nodes to create a private, "message-hopping" network that works in zero-signal zones without satellites.

Autonomous CAN Diagnostics: Instantly extracts vehicle failure data from the CAN-bus and transmits it to the nearest garage or hospital via intermediary nodes.

NLP Crisis Interface: Converts user voice into lightweight AI tokens, allowing real-time SOS communication even on low-bandwidth IoT channels.

The Multi-Objective Node Selection Equation
The Optimal Node N^* is selected by maximizing the Fitness Function (F_i):

$$N^* = \arg \max_{i \in \text{Range}} (F_i)$$

B.R.A.H.M.A

(Brain of Road Autonomous High-efficiency Mobility AI)

AI platform leverages GPS (Floating Car Data)+ CCTV feeds for adaptive signal timing (cycle, split, offset), green wave coordination, and real-time rerouting advisories.

- **B - Brain** : Central AI system that acts as the intelligent control brain for city traffic.
- **R - Road** : Monitors road conditions using GPS data and CCTV traffic feeds.
- **A - Autonomous** : Automatically manages traffic signals and routing decisions without manual intervention.
- **H - High-Efficiency** : Optimizes traffic flow, reducing congestion, delays, and emissions.
- **M - Mobility** : Handles multi-modal urban traffic such as cars, buses, autos, and two-wheelers.
- **A - AI** : Uses AI technologies

Traffic Modelling Foundation Equation :

$$q = k * v$$

q = Flow
k = Density
v = Speed (from GPS data)

S.H.I.V

((Smart Healthcare Integration & Vehicle-response)

- **ESP32 ESP-NOW Broadcast:** Hospitals send real-time bed data via peer-to-peer protocol over a range.
- **Ambulance Reception:** ESP32 on ambulance receives signals when entering range.

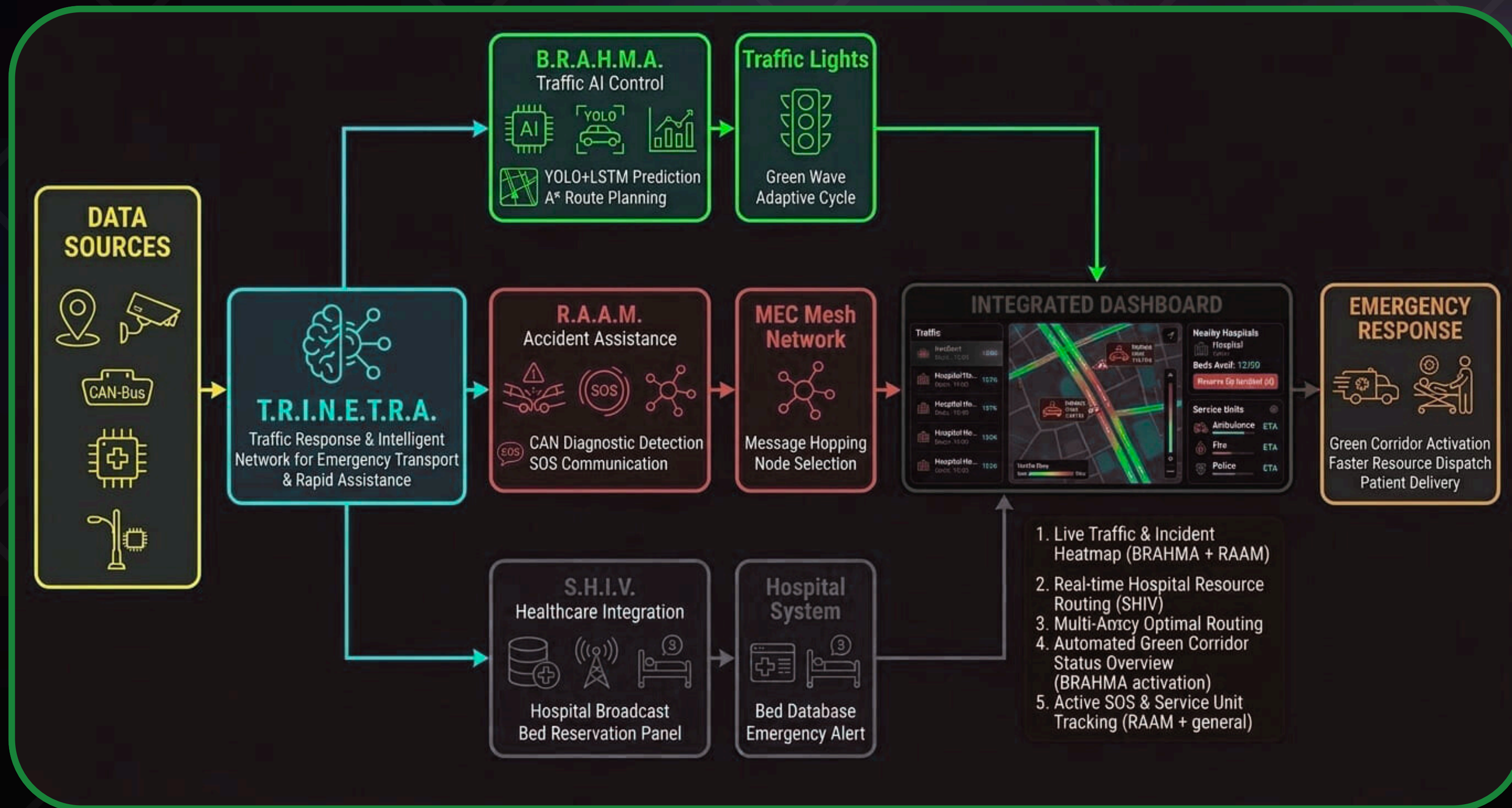
path loss:

$$\text{FSPL: } 20\log_{10}(d) + 20\log_{10}(f) + 32.4 \\ 420\log_{10}(d) + 20\log_{10}(f) + 32.44.$$

- **Native Vehicle Map Dashboard:** Live icons show nearby hospitals and bed status.
- **Instant Reservation:** Click icon to reserve; updates

$$B_{\text{available}} = B_{\text{total}} - B_{\text{booked}}.$$

ARCHITECTURE





TECHNOLOGY USED



- **Hardware** : ESP32-S3, MCP2515 CAN, IMU, IP Cameras(CCTV).
- **Software** : Ollama, Arduino IDE, OMNET++ (VEINS, ARTRIES for VANET, SUMo).
- **Language**: PyQt 6, python.
- **Protocol**: ESP-NOW ,VANET protocols (802.11.p, DSRC), CAN 2.0B.
- **Architecture**: **MEC** (Multi-access Edge Computing), **Wireless Mesh** / Ad-hoc Network (VANET), Adaptive Signal Control Technology (ASCT), **P2P**.

Technology Artitecture -

- **Networking** : Protocol Middleware (R.A.A.M.).
- **Traffic Solution** : Software Application Layer (B.R.A.H.M.A.).
- **Resource Management** : Edge Logic / P2P Layer (S.H.I.V.).



FEATURE / USP



T.R.I.N.E.T.R.A :

A multi-layered Command and Control (C2) architecture that integrates decentralized P2P communication (S.H.I.V.), edge-mesh reliability (R.A.A.M.), and centralized AI traffic flow optimization (B.R.A.H.M.A.).

USP :

The world's first "Internet-Independent" Smart City layer that uses mathematical modeling ($q=kv$) and radio physics (FSPL) to guarantee a "Green Wave" for every life-critical situation.

Crating a integrated System :

- Detection Layer (R.A.A.M) - Satellite-Independent Mesh Works when the world goes offline.
- Coordination Layer (S.H.I.V) - P2P Bed Reservatio Zero-latency hospital-ambulance sync.
- Execution Layer (B.R.A.H.M.A) - Autonomous Green Wave Mathematical destruction of congestion.



REFERENCES / LINKS

- Bengaluru: BATCS and ASTraM
- Kolkata: Project Green Light
- IISc UVH-26 dataset
- Adaptive Control and Dynamic Rerouting
- The Role of Public Policy and Inter-Agency Coordination
- IEEE : Vehicular ad-hoc network (VANET)
- On Road Vehicle Breakdown Assistance
- All about Delay-Tolerant Networking (DTN) Contributions to Future Internet

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THANK YOU